

Total marks: 20. Solve all the problems. Show your detailed work to receive the full marks.

1. Determine whether the function $y = 3x^2 \cos x - 2x^4 + 3x^2$ is an odd, even or neither. [4]

$$\text{Let, } f(x) = y = 3x^2 \cos x - 2x^4 + 3x^2.$$

$$\begin{aligned}\text{Now, } f(-x) &= 3(-x)^2 \cos(-x) - 2(-x)^4 + 3(-x)^2 \\ &= 3x^2 \cos x - 2x^4 + 3x^2\end{aligned}$$

Here, $f(x) = f(-x) = y$; so, the function $f(x)$ is even.

2. Evaluate: $\lim_{x \rightarrow \infty} \frac{\sqrt{3x^2 - 7}}{x - 6}$.

[8]

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{\frac{3x^2}{x^2} - \frac{7}{x^2}}}{\frac{x}{x} - \frac{6}{x}} \quad [\because \text{Dividing num. \& deno. by } x]$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{3 - \frac{7}{x^2}}}{1 - \frac{6}{x}}$$

$$= \frac{\sqrt{3 - \frac{7}{\infty}}}{1 - \frac{6}{\infty}}$$

$$= \frac{\sqrt{3}}{1} = \sqrt{3} \text{ (Ans)}$$

3. Find the limit of $f(x)$ at $x = 2$, where

$$f(x) = \begin{cases} x^3 - 1, & 0 \leq x < 2 \\ 2 - 2x, & x = 2 \\ \frac{1}{x-2}, & x > 2. \end{cases}$$

Solⁿ: $\lim_{x \rightarrow 2} f(x)$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 2^-} f(x) \\ &= \lim_{h \rightarrow 0} f(2-h) \\ &= \lim_{h \rightarrow 0} (2-h)^3 - 1 \\ &= (2-0)^3 - 1 \\ &= 7 \end{aligned}$$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow 2^+} f(x) \\ &= \lim_{h \rightarrow 0} f(2+h) \\ &= \lim_{h \rightarrow 0} \frac{1}{2+h-2} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \\ &= \infty \end{aligned}$$

$$\therefore \text{L.H.L} \neq \text{R.H.L}$$

$\therefore$ so the limit does not exist at $x=2$